USN. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [21EEDE741]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2024

OPERATIONS RESEARCH

Time:3 Hours **Maximum Marks:50**

Instructions:
- **i.** PART - A (Question 1-5) answer any 4 questions.
- **ii.** PART - B (Question 6-8) answer any 2 questions.
- **iii.** PART - C (Question 9) is compulsory

PART – A 05X04=20

1. a. Define the following terms i. Pay off matrix ii. Strategy iii. Two person zero sum game iv. Saddle point v. Game theory. **05**

2. a. Define transportation model, represent it in the form of general matrix. **05**

3. a. Explain the steps followed in Hungarian method. **05**

4. a. Explain the steps involved in Fulkerson's rule (i - j) rule. **05**

5. a. Explain the maximin – minimax principle of game theory. How is the game theory problems classified? **05**

PART - B 09X02=18

6. a. Explain some of the applications of operations research to engineering and managerial problems. **06**

b. Explain the basic steps in project management. **03**

7. a. A dairy firm has three plants located in a state. The daily milk production at each plant is as follows: **09**

Plant 1 : 6 million litres, Plant 2 : 1 million litres, and Plant 3 : 10 million litres

Each day, the firm must fulfil the needs of its four distribution centres.

The minimum requirement of each centre is as follows:

Distribution centre 1 : 7 million litres, Distribution centre 2 : 5 million litres,

Distribution centre 3 : 3 million litres, and Distribution centre 4 : 2 million litres

Cost (in hundreds of rupees) of shipping one million litre from each plant to each distribution centre is given in the following table:

		Distribution Centre			
		D_1	D_2	D_3	D_4
Plant	P_1	2	3	11	7
	P_2	1	0	6	1
	P_3	5	8	15	9

Find the initial basic feasible solution for given problem by using following methods:

- **i.** North-west corner rule
- **ii.** Least cost method
- **iii.** Vogel's approximation method

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

8. a. Solve the travelling salesman problem given by the following data.

C12 = 20, C13 = 4, C14 = 10, C23 = 5, C34 = 6, C25 = 10, C35 = 6, C45 = 20 where Cij = Cji and there is no route between cities I and j if the value for Cij is not shown. **06**

b. Differentiate between transportation model and assignment model. **03**

PART - C

01X12=12

9. a. Two players A and B put a coin each simultaneously on the table without showing each other. A wins Rs.8/- when both coins show head and Rs.1/- when both coins are tail. Player B wins Rs.3/- when the coins do not match. Prepare a payoff matrix and determine the best strategy of each player and value of the game.

Player A \ Player B	B1	B2
A1	8	-3
A2	-3	1
A3	8	1

b. Use the graphical method to reduce the following game and hence solve. **06**

$$\begin{bmatrix} 3 & -3 & 4 \\ -1 & 1 & -3 \end{bmatrix}$$